Higher National Diploma in Engineering (C \ E \ M BSE)

First Year, First Semester Examination -- 2019

MA 1101: Engineering Mathematics

Question 1

a) Find the **Eigen values** and the corresponding **Eigen vectors** for the following matrix:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$$

(15 Marks)

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b) Write down the **quadratic form** corresponding to the matrix:

$$A = \begin{pmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 1 & 1 \end{pmatrix}$$

(6 Marks)

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c) Write down the **matrix** of the quadratic form:

$${x_{1}}^{2}+{2x_{2}}^{2}+{2{x_{3}}^{2}-2x_{1}x_{2}-2x_{3}x_{2}$$
(4 Marks)

(Total 25 Marks)

Question 2

a) Show that the following series is **convergence if** $x < 1$ **and divergence if** $x > 1$:

$$1 + \frac{2x}{5} + \frac{6x^2}{9} + \frac{14x^3}{17} + \dots$$

[The full expression of the series is not clearly stated, but the question appears to relate to the series $\sum_{n=1}^{\infty}(\frac{2^n-2}{2^n+1})x^{n-1}$ based on proximity in the text. Assuming the series is $\sum_{n=1}^{\infty}(\frac{2^n-2}{2^n+1})x^{n-1}$] (15 Marks)

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b) Test for **convergence of the series**:

$$\sum_{n=1}^{\infty}(\frac{2n^3+5}{4n^5+1})$$

(10 Marks)

(Total 25 Marks)

Question 3

a) Evaluate $\lim_{x\to 0} \frac{\log x}{\cot x}$. (05 Marks)

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b) If $Y = A\cos(\log x) + B\sin(\log x)$ Prove that:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$$

(05 Marks)

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c) If $u = \sin^{-1}(\frac{x+y}{\sqrt{x}+\sqrt{y}})$ Using **Euler's theorem** on homogeneous function, Show that:

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{\tan u}{2}$$

(06 Marks)

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d) If $U = \log(x^3 + y^3 + z^3 - 3xyz)$ Show that:

$$(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z})^2 U = \frac{-9}{(x+y+z)^2}$$

(09 Marks)

(Total 25 Marks)

Question 4

a) If $U = \frac{yz}{x}, V = \frac{xz}{y}, W = \frac{xy}{z}$ Show that:

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$$

(15 Marks)

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b) **Expand e^x in powers of x and y up to five terms.** (10 Marks)

(Total 25 Marks)

Question 5

a) Evaluate the **double integral** $\iint_D (x+2y)\,dx\,dy$ where D is the region bounded by the curve $y = 2x^2$ and $y = 1 + x^2$. (12 Marks)

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b) Find the **volume** bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$, $z = 0$. (13 Marks)

(Total 25 Marks)